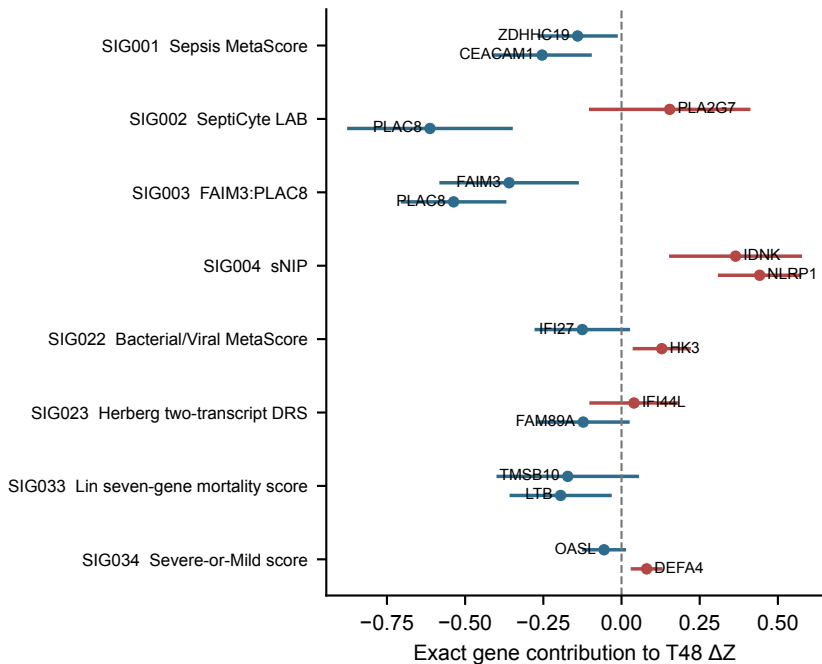
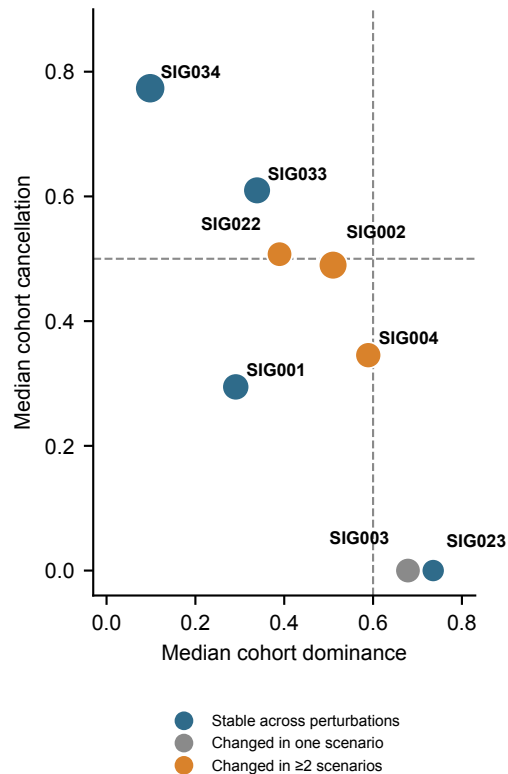


Gene-contribution patterns at T48

a Two largest pooled mathematical contributions



b Continuous metrics and label sensitivity



Continuous metrics are primary. Each point is the median of cohort-specific patient medians from the same primary-independent set used for score synthesis. Colors summarize label sensitivity under uniform $\pm 10\%$ and $\pm 20\%$ threshold perturbations; contributions are mathematical, not causal.